

four new automotive grade devices in a compact eSMP SMF package. These 1A and 2A rectifiers, designed

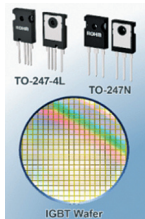


for industrial and automotive use, combine low junction capacitance, fast 45ns recovery time, and minimal reverse recovery charge, improving efficiency and reducing switching losses. Ideal for snubber, clamp, and freewheeling diodes, they support temperatures up to +175°C and meet RoHS and AEC-Q101 standards. Samples are available immediately.

Vishay Intertechnology, Inc
<https://www.vishay.com>

4th Generation IGBTs

ROHM has introduced AEC-Q101-qualified 1200V 4th Generation IGBTs for automotive and industrial applications requiring high efficiency and voltage handling. With a high withstand voltage of 1200V and enhanced current interruption capabilities,



these IGBTs improve energy efficiency, ease cooling, and optimise design. Kelvin emitter terminal design reduces losses by up to 35%, enhancing efficiency in inverter applications.

ROHM Semiconductor
<https://www.rohm.com>

USB power delivery controllers

The new AP33771C and AP33772S controllers are ideal for battery-powered devices like routers, wireless speakers, power banks, and power tools with USB Type-C power sourcing. They



offer faster, more efficient charging than competitors by supporting USB PD 3.1 EPR up to 140W and adjustable voltage supply up to 28V. Both

controllers include smart firmware for practical applications, such as LED indicators and moisture detection, with robust system protection and cost-saving NMOS FET support, reducing component requirements.

Diodes Incorporated
<https://www.diodes.com>

Hydrogen sensors

Metis Engineering's new cell guard with hydrogen sensor provides crucial hydrogen leak detection, essential in high-risk industries such as transport, marine, aviation, and industrial operations where hydrogen's flam-

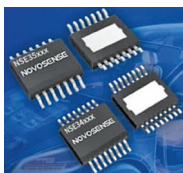


mability and explosive nature pose safety challenges. The sensor also monitors fuel cells and battery packs, detecting hazardous electrolysis gas production. Built for temperature range of -40°C to +85°C with ultra-low power usage, it offers real-time alerts via CAN interface, performing effectively in high-VOC environments without warm-up delays.

Metis Engineering
<https://metisengineering.com>

High-side switches

NOVOSENSE Microelectronics has introduced high-side switches designed for automotive body control modules and zone control units,



handling resistive, inductive, halogen, and capacitive loads. The NSE34 and NSE35 series offer 26 single-, dual-, and quad-channel devices, managing load currents from 11A to below 2A with resistance between 8mΩ and 140mΩ. With AEC-Q100-012 certification and robust protections like overcurrent, short-circuit, and reverse polarity safeguards, these switches suit automotive OEMs, component makers, and aftermarket

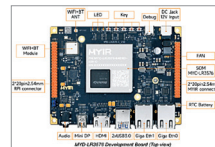
suppliers.

NOVOSENSE Microelectronics
<https://www.novosns.com>

EMBEDDED

Octa-core 4K video processor

MYIR's MYC-LR3576 system-on-module (SoM), powered by the Rockchip RK3576, is tailored for AIoT, edge computing, and industrial applica-



tions, offering a balance of high performance and compact design.

Equipped with an octa-core processor and a 6 TOPS NPU, it supports 4K video processing. Flexible connectivity includes dual gigabit Ethernet, USB 3.2, and PCIe, catering to smart devices and multimedia needs. The compact SoM provides many storage options, with Debian 12 and Linux 6.1 support.

MYIR
<https://www.myrirtech.com/default.asp>

RGB-IR camera

The camera, featuring onsemi's AR0830 sensor, captures high-quality 4K images in both RGB and infrared light, suitable for applications like



biometric access, in-cabin monitoring, crop health assessment, image-guided surgeries, and smart patient monitoring.

By streaming RGB-IR frames simultaneously, it provides clear images under various lighting conditions. A unique RGB-IR separation algorithm enables single-sensor processing, reducing the need for multiple sensors, which minimises latency and costs.

e-con Systems
<https://www.e-consystems.com/default.asp>

SoM for developing industrial HMI and gateways

Human-machine interfaces (HMIs) and gateways are crucial for connecting